



Fix the Code

Quick facilitation notes & answer key

Goal

Students debug a program: run it, see Bit miss the target, find the one wrong block, fix it, and run again to confirm. Reading existing code, finding the error, and altering it so the outcome is correct is C3.2 in action — and a first, structured debugging routine.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 with a marker on 4 (the target) so students can walk the buggy code and feel Bit go the wrong way.

Run it (about 20 min)

1. I DO: run the buggy code — 'start 1... forward 2 to 3... back 1 to 2. Bit is on 2, but the target is 4. Something is wrong.'
2. WE DO: Exercise 1 — the buggy code lands on 2 (not 4). Exercise 2 — the back 1 block moves Bit away from 4; circle it.
3. YOU DO: Exercise 3 — replace the wrong block with forward 1; the code becomes start 1, forward 2, forward 1 and reaches 4.
4. Connect: 'Run, find the wrong block, fix it, run again — that is how coders debug.'

Answer key (modelled by the run-gate)

Ex 1 — buggy code: 1 → forward 2 → 3 → back 1 → 2. Bit lands on 2, not the target 4.

Ex 2 — the wrong block is back 1: it sends Bit backward, away from 4.

Ex 3 — fix it to forward 1: 1 → 3 → 4. Bit lands on the target, 4.

Why this is the bug: the first two blocks bring Bit to 3, one step short of 4; a forward 1 finishes the job, but a back 1 undoes progress and lands on 2.

Watch for & differentiate

Common slip: blaming the start or the forward 2 — have students run step by step and watch exactly where Bit turns away from 4.

Simplify: walk the buggy code on the floor path, then swap the last card to forward 1 and re-walk.

Extend: ask for a DIFFERENT single-block fix that also reaches 4 — e.g. change forward 2 to forward 4 and keep back 1: 1 → 5 → 4. Have students test their idea by running it.

Success indicator: the child lands the buggy code on 2, identifies back 1 as wrong, and fixes it to forward 1 to reach 4.